

RAMYA AKULA, PH.D.

AI Researcher & Applied Machine Learning Leader

San Jose, CA | ramya.akula01@gmail.com | +1 407-721-7814 | [LinkedIn](#) | [Google Scholar](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Ph.D.-trained AI researcher and educator with 10+ years across applied machine learning, NLP, computational biology, industry engineering, and interdisciplinary research. Experienced in turning ambiguous technical problems into reproducible workflows, interpretable models, and decision-ready insights. 2022 DARPA Forward Riser, U.S. patent holder, and author of peer-reviewed research with 267 citations. Strong communicator with experience teaching graduate AI courses, mentoring teams, supporting customer discovery, and collaborating with scientific, engineering, and business stakeholders.

CORE STRENGTHS

Trustworthy AI Interpretability, evaluation, reliability, human oversight	NLP & Generative AI LLMs, transformers, summarization, semantic evaluation	Computer Vision Image/video analytics concepts, model evaluation, OpenCV
Computational Biology Single-cell RNA-seq, transcriptomics, biological interpretation	Applied AI Strategy Problem framing, stakeholder discovery, roadmaps, success metrics	Research Leadership Cross-functional collaboration, mentoring, scientific communication

PROFESSIONAL EXPERIENCE

Postdoctoral Scholar - Computational Biology & Machine Learning The Ohio State University, Department of Internal Medicine (Pulmonary) Columbus, OH	2024-2025
- Developed reproducible pipelines to integrate multi-cohort single-cell RNA-sequencing datasets, including quality control, Harmony-based batch correction, cell-type annotation, differential expression, and pathway-level interpretation. - Analyzed transcriptomic signatures of lung aging, cellular senescence, apoptosis, donor type, and chronic lung injury across human and mouse cohorts. - Partnered with pulmonary scientists to translate biological questions into rigorous analytical plans, interpretable visualizations, and publication-ready findings. - Contributed to a 2026 peer-reviewed publication in The Journals of Gerontology.	
Lecturer - Computer Science & M.S. FinTech University of Central Florida Orlando, FL	2023-2024
- Designed and taught graduate and undergraduate courses in artificial intelligence, NLP, machine learning, data science, algorithms, blockchain, smart contracts, and programming. - Mentored approximately 80 students across applied AI, data-science, research, and software-development projects. - Translated complex technical concepts for audiences with varied backgrounds and advised student teams on project scope, methodology, execution, and communication.	
Graduate Research Assistant - Trustworthy AI & NLP University of Central Florida, Complex Adaptive Systems Laboratory Orlando, FL Advisor: Dr. Ivan Garibay	2018-2022
- Developed an interpretable multi-head self-attention architecture for sarcasm detection; the work received a U.S. patent and DARPA Forward Riser recognition. - Proposed SPEED Score, a sentence-embedding-based metric for evaluating abstractive and extractive summarization. - Researched explainable AI, figurative-language understanding, toxic-language detection, semantic evaluation, and AI for social good under DARPA-supported programs. - Served as NSF I-Corps Entrepreneurial Lead, conducting customer discovery and commercialization activities for an AI-assisted qualitative-analysis platform.	
Project Engineer - Airbus A350 Program Umlaut GmbH (formerly P3 Group), Client: Airbus GmbH Hamburg, Germany	2015-2017
- Supported the Airbus A350 product-structure transformation program through stakeholder meetings, requirements gathering, estimation, milestone planning, and engineering coordination. - Developed tooling to reduce computational overhead in product-structure workflows and improve execution efficiency.	

ADDITIONAL INDUSTRY & RESEARCH EXPERIENCE

Earlier roles: Software Quality Assurance Analyst (Roche client); Graduate Research Assistant (RPTU and DFKI, Germany); Software Engineer (Osmania University); Software Developer (UnitedHealthCare client).

SELECTED RESEARCH & INNOVATION

- U.S. Patent 12,210,833 (2025): Interpretable Multi-Head Self-Attention Architecture for Sarcasm Detection.
- DARPA Forward Riser (2022): Recognized for innovative research in interpretable machine learning for figurative-language understanding.
- NSF I-Corps Entrepreneurial Lead (2019): Led customer discovery and commercialization activities for an AI-assisted qualitative-analysis platform.
- First Place, Startup Weekend Orlando (2021): Contributed to an AI-powered online youth-safety platform.
- Research featured by NBC News, IEEE Spectrum, Los Angeles Times, Defense One, ScienceDaily, and ABC Australia.

SELECTED PUBLICATIONS

- Akula, R., et al. “Transcriptomic shifts in human lungs donated after brain and circulatory death and severe Obliterative Bronchiolitis.” The Journals of Gerontology, 2026.
- Akula, R., & Garibay, I. “SPEED Score: Sentence Pair Embeddings Based Evaluation Metric for Abstractive and Extractive Summarization.” LREC, 2022.
- Akula, R., & Garibay, I. “Interpretable Multi-Head Self-Attention Architecture for Sarcasm Detection in Social Media.” Entropy, 2021.
- Akula, R., & Garibay, I. “Explainable Detection of Sarcasm in Social Media.” ACL Workshop, 2021.

TECHNICAL EXPERTISE

Natural Language Processing & Generative AI: Large language models, transformer architectures, conversational AI, text summarization, semantic similarity, figurative-language understanding, toxicity detection, prompt and model evaluation
Computer Vision & Video Analytics: Image classification, object detection, object tracking, segmentation, activity/event recognition, video-processing workflows, OpenCV, model evaluation
Trustworthy AI & Machine Learning: Interpretability, robustness, uncertainty, bias evaluation, human-AI collaboration, deep learning, transfer learning, representation learning
Computational Biology: Single-cell RNA sequencing, Scanpy, Seurat, Harmony, differential expression, cell annotation, gene-set enrichment, transcriptomics
Data & Scientific Computing: Python, R, PyTorch, TensorFlow, Jupyter, Git, Linux, Docker, high-performance computing, reproducible workflows

EDUCATION

Ph.D. in Computer Science University of Central Florida Orlando, FL Advisor: Dr. Ivan Garibay	2018-2022
M.S. in Computer Science Technical University of Kaiserslautern Germany Advisor: Dr. Hans Hagen	2013-2015
B.Tech. in Computer Science and Engineering Jawaharlal Nehru Technological University India	2006-2010